# Tossed Coins More Likely to Land Same Side Up, Say Researchers

A coin toss isn't as fair as everybody thought, say scientists, after 350,000 flips.

The Physics arXiv Blog
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When Italy's national soccer team took on the Soviet Union in the semi-finals of the European Championship in 1968, the score was 0-0 at the end of the game and a further period of extra time failed to separate the two sides.

The rules of the time dictated that in these circumstances, the game should be decided by the toss of a coin. Italy's captain Giacinto Facchetti called correctly, and Italy proceeded to the final, which they eventually won.

A coin toss is often considered the fairest way to settle an otherwise intractable question. The process involves using your thumb to launch a coin into the air and then catching it again. The chances of it landing on one side or the other — heads or tails — are 50:50, which is what makes the process fair. Or so everyone had thought.

Quiz

What is the only planet in our solar system that spins clockwise?

Jupiter

Mercury

Venus

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## Fair Toss

Enter František Bartoš, at the University of Amsterdam in the Netherlands and several dozen collaborators, who have collected the results of 350,757 coin flips and conclude that the outcomes are not as had been thought. “Fair coins tend to land on the same side they started,” they conclude, saying the result confirms a 15-year-old theory that the act of coin tossing is not quite as fair as might be expected.

First some background. Bartoš and co say that in the standard model of coin tossing, a coin flip obeys the laws of Newtonian physics in a transparent fashion with the randomness originating in tiny differences in the starting conditions of the toss.

For example, the tosser might use slightly different levels of force to flip the coin, this force might be exerted at different points on the coin, leading to tosses that spin faster or remain in the air longer. All this should lead to a 50:50 probability that the coin lands heads or tails up. And indeed, the evidence is that this is true.

But in 2007, the statistician and magician, **Persi Diaconis with a couple of colleagues, developed a more detailed model** that predicted an additional effect. In their model, the tossed coin does not spin perfectly around an unchanging axis. Instead, the tosser can introduce a wobble or precession — a change in direction in this axis of spinning. “Precession causes the coin to spend more time in the air with the initial side facing up,” say Bartoš and co. “Consequently, the coin has a higher chance of landing on the same side as it started.”

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The effect is small. Diaconis and co calculated that it should be about 0.51 — in other words, the coin should land on the same side as it started 51 percent of the time. And they took high-speed videos of flipped coins to show this wobble.

The trouble is that proving such a subtle effect requires a large number of tosses. And until now, reliable evidence had either not recorded the starting position of the coin or did not record enough events to be statistically significant.

That's where Bartoš and co come in. This group of 48 people tossed coins from 46 different countries a total number of 357,757 times, while recording both the starting and final configurations.

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## Wobbly Tossing

How tossers introduce more bias is not yet clear. “Future work may attempt to verify whether "wobbly tossers" show a more pronounced same-side bias than "stable tossers."” That will require significantly more tosses with high-speed video analysis of each.

The data also suggests that coins are otherwise fair, with no evidence of a bias in favor of heads or tails.

That’s interesting work raising the important question of whether it is possible to exploit this bias. Bartoš and co explore this question using the following betting scenario. “If you bet a dollar on the outcome of a coin toss (i.e., paying 1 dollar to enter, and winning either 0 or 2 dollars depending on the outcome) and repeat the bet 1,000 times, knowing the starting position of the coin toss would earn you 19 dollars on average.”

That may not sound like much, but the researchers point out that it is more than the $5 a casino would expect to make against an optimal strategy in 6-deck blackjack. But less than the $27 it might make in single-zero roulette.

That points the researchers towards an actionable conclusion: “These considerations lead us to suggest that when coin flips are used for high-stakes decision-making, the starting position of the coin is best concealed”.

Which leaves only the question of how Giacinto Facchetti made his call in the semi-final of the European soccer championship in 1968.

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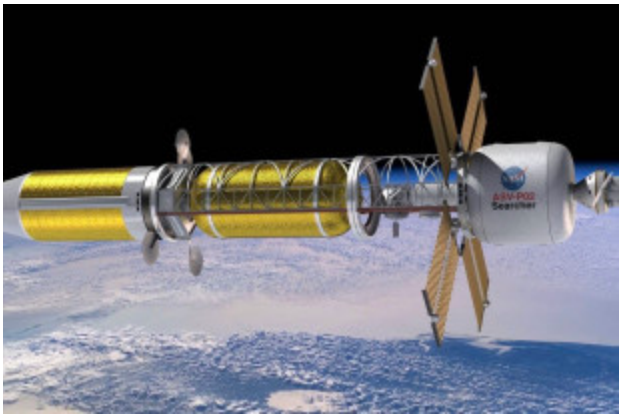
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